

Hamza Madi

Computer Science and IT

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EDUCATION

Feb 2022-
Dec 2025

University Of KwaZulu-Natal, Durban BSc in Computer Science and IT

Computer Science graduate and **Artificial Intelligent intern** with experience in Python, data analysis, and full-stack development. Built database-driven applications and supported students as a university demonstrator, strengthening both technical and communication skills.

Relevant Coursework

Computer Science:

Data Structures, Algorithms, Programming Languages, Web Programming, Machine Learning

Informatics:

Computer Networks, Databases, Client-Side Development, Systems Design and Analysis

Feb 2026-
Progress

IBM Data Engineering Professional Certificate (In Progress) — IBM

Learning data pipelines, ETL processes, SQL, Python, and cloud-based data engineering

PROFESSIONAL EXPERIENCE

Jan 2026-
June 2026

AI and Data Science intern –Hulamin

- Gained hands-on experience in artificial intelligence and data science concepts, including data processing, analysis, and model development.
- Worked with structured and semi-structured datasets to extract insights and support decision-making.
- Applied Python for data analysis and basic machine learning tasks.
- Collaborated with team members on real-world data-driven projects and problem-solving tasks.
- Developed an understanding of data pipelines, data cleaning, and visualization techniques.

April 2025-
May 2025

AI Consulting Volunteer, ITRelated

- Volunteered on AI-focused initiatives as a final-year Computer Science student
- Researched and presented recommendations on how Artificial Intelligence could be implemented within the organisation through a formal report and presentation..

July 2024-
Nov 2025

Computer Science Demonstrator, University of KwaZulu-Natal

- Facilitated demonstrations and provided academic support in Java programming, Object-Oriented Programming (OOP), Unix/Linux systems, and computer networking concepts.
- Guided students through coding exercises, troubleshooting, network simulations, and real-time traffic analysis to strengthen practical problem-solving skills.
- Applied data distribution techniques to analyse network load, resource utilization, and system performance.
- Enhanced communication, mentoring, and technical support skills through hands-on student engagement and collaborative learning environment

Personal Projects

Manufacturing Defect Prediction System:

- Built a machine learning model to predict defective products using manufacturing data (cost, machine downtime, productivity, energy usage).
- Performed data cleaning, feature engineering, and applied classification algorithms to identify key factors contributing to defects.

Technologies: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Power BI

Clinic Management System Web Application :

- Designed and implemented clinic management solution for patient records, and appointments.
- Created database-driven architecture with reporting dashboards

Technologies: .NET (C#), MySQL, HTML, CSS, JavaScript, Power BI

ProNearby Platform:

- Built location-based service platform connecting users with nearby professionals
- Developed backend using Flask with PostgreSQL database integration
- Implemented map-based functionality using Mapbox API

Technologies: Python, Flask, PostgreSQL, JavaScript, Mapbox

MzansiBuilds – Developer Collaboration Platform:

- Designed and developed a full-stack platform for developers to create projects, share progress, and collaborate in real time.
- Implemented authentication, project management, collaboration requests, project feeds, and a celebration wall for completed projects.

Technologies: React.js, Flask (Python), PostgreSQL, JavaScript, HTML, CSS

- Explore my projects on **GitHub**:
[github.com/lionhamza](<https://github.com/lionhamza>)

CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals (AZ-900)
- Microsoft Certified: Azure AI Fundamentals (AI-900)
- Microsoft Certified: Azure Data Fundamentals (DP-900)

TECHNICAL SKILLS

- ❖ **Programming Languages:** Java, Python, C++, C#, SQL.
- ❖ **Databases:** PostgreSQL, MySQL
- ❖ **Backend & Web:** Flask, REST APIs, HTML, CSS, JavaScript
- ❖ **Data & Analysis Tools:** Debugging, testing strategies, full software development lifecycle. MS Excel (Advanced Formulas, Pivot Tables), Power BI

SKILLS & ABILITIES

- Strong problem-solving and debugging skills with attention to detail.
- Experienced in building responsive, scalable, and user-focused applications.
- Effective communicator and collaborator in cross-functional teams.
- Quick learner, adaptable to new tools, frameworks, and emerging technologies.
- Skilled in the full software development lifecycle, from design to deployment and software testing.